DATA ANALYTICS - PHASE 3 REPORT

Name: VENI PRASANNA

Track: Data Analytics

Phase: Phase 3 – Money and Insight

Project Name: PulseIQ – Workforce and Business Analytics Dashboard

GitHub Repository: https://github.com/VENIPRASANNA/growmar_pulseiq

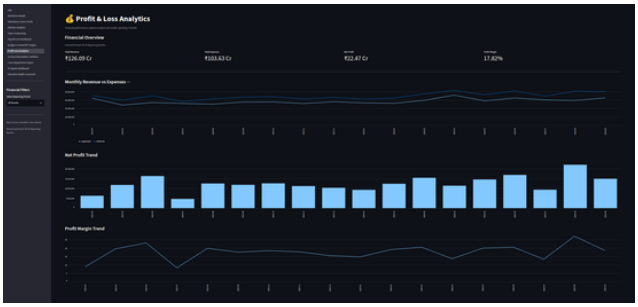
Phase 3 Overview

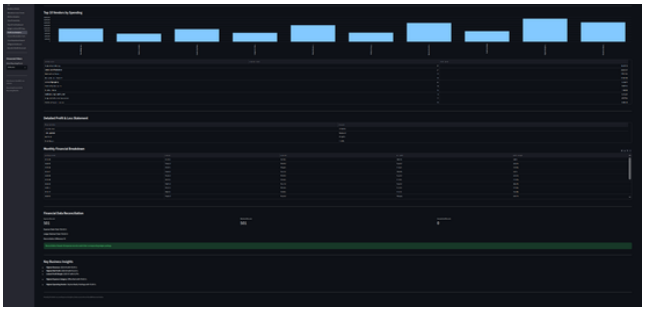
Phase 3 focused on developing advanced analytical dashboards to support business monitoring and management decision-making. Five dashboards were developed using Python, Streamlit, Pandas, Plotly and DuckDB. The dashboards cover financial performance, invoice management, cross-department analysis, AI-generated signals and executive-level business health.

Tasks Completed with Screenshot Evidence

Task 11 – Profit & Loss and Expense Analytics

- Analysed revenue, expenses, net profit and operating margin.
- Developed financial visualizations and expense breakdowns.
- Validated 501 expense records with zero unmatched records and ₹0 reconciliation difference.





Task 12 – Invoices, Receivables and Cash Flow

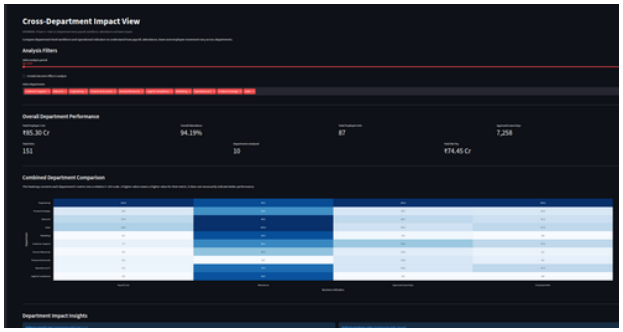
- Analysed 743 invoices, open receivables and overdue payments.
- Implemented invoice ageing and cash-flow analysis.
- Identified ₹13.94 Cr open receivables and ₹2.71 Cr overdue receivables.





Task 13 – Cross-Department Impact Analysis

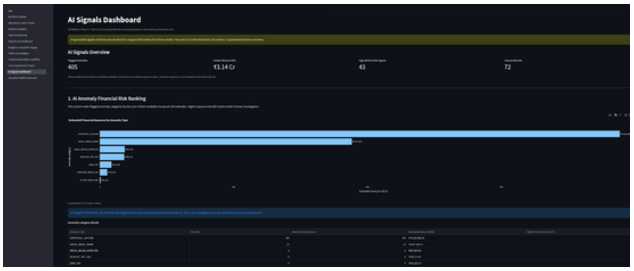
- Compared workforce and financial metrics across departments.
- Generated 198 department-month summary records covering 11 departments and 18 months.
- Developed comparison charts and heatmaps.

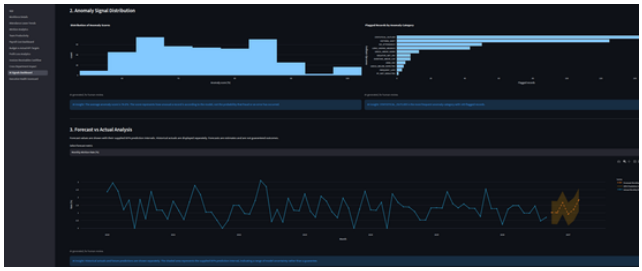




Task 14 – AI Signals Dashboard

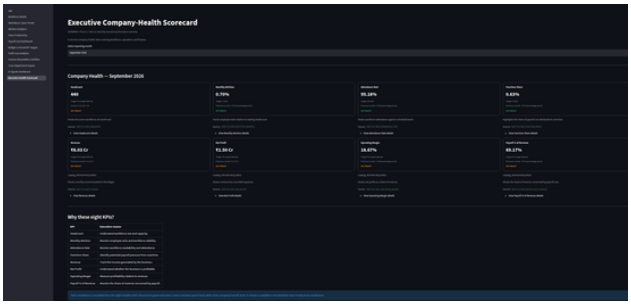
- Analysed 405 flagged anomaly records.
- Identified approximately ₹3.14 Cr in known estimated financial exposure.
- Implemented anomaly ranking, forecast analysis and department-level attrition risk visualization.





Task 15 – Executive Company-Health Scorecard

- Included actual values, targets, previous-month trends and status indicators.
- Connected the KPIs to their detailed dashboards.



Mini Challenge

- Management Question: What financial risks should management monitor?
- Revenue: ₹126.09 Cr.
- Expenses: ₹103.63 Cr.
- Net Profit: ₹22.47 Cr.
- Overdue Receivables: ₹2.71 Cr.
- Known AI-detected financial exposure: ₹3.14 Cr.
- These dashboard metrics help management understand profitability, collection risks and unusual financial patterns.

Problems Faced and Solutions

- Data mismatches: Validated records and documented discrepancies without changing source values.
- Missing or invalid data: Used data cleaning and quarantine checks to handle incorrect records.
- Calculation differences: Reconciled dashboard metrics with reference values.
- Dashboard errors: Fixed Python and Streamlit issues through testing and debugging.
- AI prediction limitations: Clearly labelled forecasts and anomaly results for human review.

Learning and Challenges

- Learned to combine multiple datasets at the correct level of detail.
- Improved skills in DuckDB, Python, Streamlit and data visualisation.
- Faced data mismatches and resolved them through validation and reconciliation.
- Learned to present business insights through interactive dashboards.

Next Phase Plan

- Improve dashboard performance and design.
- Perform deeper business analysis.
- Strengthen data validation and reporting.
- Prepare the final project documentation.

Conclusion:

- Successfully completed Phase 3 by developing five interactive dashboards and validating the key business metrics.